

Integrative Core Case Summer 2025

Georgia Plant Care Company (GPC)

Team 2

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Executive Summary

Georgia Plant Care Company (GPC) is preparing to launch PlantLuv, its first-ever consumer gardening product. This strategic expansion into a competitive and unfamiliar market presents challenges across operations, marketing, and finance—but our analysis supports a clear path forward for a successful, scalable, and financially sound entry.

Cultivating the Ideal Sales Team

Staying true to its roots, GPC will continue to foster and exemplify a relationship-based culture, focusing on nurturing connections with their client-base with an added emphasis on internet sources leads. An opportunity for growth arose when analyzing Georgia Plant Care Co.'s current sales team precedent and innovation development incentives system. Through careful reconciliation and analysis, we at the KAKAV Group have recommended a tailored organizational change strategy that is sure to invite innovation and care representative of not only the PlantLuv initiative, but also Georgia Plant Care Co. as a whole.

We took the liberty of creating a job description that measures all relevant Knowledge, Skills, Abilities, and Other Characteristics (KSAOs) for a new type of salesperson specifically designed to sell the PlantLuv system to your target segments, Gardening Elders and Garden Proud. In addition to this job description, there is a comprehensive list of 3 behavioral-based questions with a 5-Point Rating Scale to help you, the executives at GPC, to home in on the best possible candidate from a selection of already qualified candidates.

Although there is immense potential for upside in pursuing this innovative PlantLuv initiative, it isn't without its threats. From a company management perspective, the culture built so pervasively on fostering connection and treating customers like family exists due to the composition of GPC's current sales force. GPC's sales force is comprised of elderly individuals who embody strong, traditional values and this translates directly into a self-serving company culture, however, these strong values tend to narrow the opportunity for growth and innovation. By implementing some change and innovation strategies and rituals, we ensure that your sales

force will be able to evolve and adapt to any future innovations Georgia Plant Care Co. manufactures and will retain the core competencies of connections the company was founded in.

Winning in a Crowded Market

GPC's strategic move in entering the consumer gardening market poses significant challenges. Not only does GPC have no consumer marketing experience, but the consumer market is very competitive, with 75% being dominated by Royal and Carebest. To tackle these issues, GPC must position itself as a premium solution for serious gardening enthusiasts with high purchasing power and large garden. Using premium pricing and tailored integrated marketing communications, GPC can build trust and reinforce perceived value to consumers and differentiate between competitors. Given its lack of direct-to-consumer infrastructure, GPC should pursue an indirect distribution strategy through gardening specialty retailers. This channel allows the company to leverage existing consumer trust and retail expertise, enabling faster, lower-risk market entry while focusing internal resources on product refinement and brand development.

The consumer gardening market has been increasing, especially during COVID, showcasing good opportunities to enter the market. Our targeted segments, Gardening Elders and Garden Proud, have a profit potential of \$117 million and \$ 243 million respectively. Being able to capture these segments will lead to high profits and satisfy stockholders. However, failure to effectively communicate PlantLuv's unique value, build brand trust, or capture early adopters could lead to low adoption rates and project failure. If the product's differentiation is not clearly reinforced, competitors may easily replicate key features, diminishing PlantLuv's long-term value. Furthermore, while partnering with specialty gardening stores allows fast market penetration, Georgia Plant Care's control over customer experience, branding, and positioning can be limited. The company must proactively collaborate with these retail partners to ensure that PlantLuv's premium positioning and brand message are consistently delivered.

Linking Operational Excellence to Customer Satisfaction

To ensure a smooth and cost-effective launch, the operations team recommends investing in a machine-paced production system. Although this requires a higher upfront investment, it offers long-term cost efficiency and

scalability, particularly as demand is expected to grow by 10% annually. A comprehensive project schedule was developed, and the timeline was reduced from 36 to 29 weeks through targeted project crashing, allowing GPC to stay on track for the planned launch. Additionally, supplier analysis led to the recommendation of continuing with existing partner M2, whose short lead times and strong relationship with GPC support lower inventory costs and reduced risk.

Beyond ensuring a timely launch and cost-efficient setup, operations will play a critical role in delivering consistent quality and supporting long term success. As GPC enters the consumer market, adopting a more proactive quality approach will be essential. Lean and Six Sigma methods like 5S, DMAIC, and FMEA can help reduce variation, prevent errors, and build reliability into each step of the process. This shift from traditional quality control to a broader quality assurance mindset will be key for maintaining brand trust, especially with a tech-based, personalized product like PlantLuv. Additionally, operations should not just support strategy but help drive it—enabling innovation, managing complexity, and ensuring that customer expectations are consistently met as the company grows.

Financial Validation of Strategic Direction

From a financial perspective, the PlantLuv project shows strong viability, with a baseline NPV of \$10.32 million and IRR of 180%, even under conservative assumptions. Sensitivity analysis across different demand scenarios provided by the operations team still produced NPVs well above zero (e.g., \$4M+ in the pessimistic case), reinforcing confidence in the project's long-term profitability despite operational uncertainty. Our team used a carefully calibrated CAPM-based discount rate, accounting for project-specific risk and liquidity concerns to ensure realistic valuation. While the initial capital requirement exceeds \$670K, the project's positive and growing cash flows throughout the horizon demonstrate that the long-term payoff justifies the investment. Additionally, the financial model accounts for key risks, such as the erosion of legacy product sales, through conservative forecasting and premium adjustments for liquidity risk. Importantly, the success of the project is directly tied to operational decisions, such as automation and machinery investment, highlighting a strong cross-functional dependency.

Additionally, we understand that the business world is full of ever-changing situations. It is vital that all business projects are sustainable in a variety of circumstances. To see if the PlantLuv system was viable to invest in terms of its stability with changing market conditions, we conducted a secondary analysis with various adaptations. This analysis concluded that the project still had impressive financial returns, however, these returns showed a drastic decrease from the initial analysis. Consequently, we recommend that Georgia Plant Care Company continues with the project due to its positive NPV and IRR on the condition that management is aware of its sensitivity to change. With proper monitoring, we believe the project has great potential but only if watched closely.

Final Recommendation

After a comprehensive and holistic assessment of Georgia Plant Care Co.'s management, marketing, operations, and financial sectors, we at the KAKAV Group recommend that GPC go ahead and invest in the PlantLuv project. Although the initial financial investments may seem steep and the initiative will require continued restructuring and ongoing expenses, the long-term benefits present a compelling case. Our analysis shows that the growing consumer demand, scalable operational strategy, and premium market positioning all work together to create a strong foundation for success. The PlantLuv system not only introduces a new revenue stream but also sets the stage for GPC to evolve into a modern, customer-focused business capable of innovation and growth in the consumer space. With thoughtful execution, continued cross-functional alignment, and close monitoring of performance and market trends, PlantLuv has the potential to generate significant financial returns and expand GPC's presence into homes and gardens across the country.

Adaptive Leadership: Success in PlantLuv Integrative

Leadership is the who behind a company's daily operations. The priorities and employees established by the leadership of GPC are vital to the company's daily and long-term successes. Even with GPC's prior success, this new project requires adaptations to their original leadership program, while still honoring their company identity.

Mission and Culture

Company Identity Identification

The culture that best aligns with both GPC and the PlantLuv launch is a Clan Culture, and this should remain as their intention. This company was founded on family values. As Len described, “those plants had been so precious to [his] botanist father,” and that legacy ignites GPC's operations. Their past success has come from targeting niche markets and creating strong customer relationships. Their first sales manager, Bob, “treats his customers like family” which is not just a sales tactic, but the ingrained culture of the Georgia Plant Care Company.

A Clan Culture emphasizes trust, loyalty, mentorship, and strong personal connections, all of which align with what the PlantLuv system is built to do. This is not just another tech product, but a way for gardening enthusiasts to connect with the practice and others in a deeper, emotional state. This goal fits right in with GPS's traditional culture of care for their product, and customers. There is no need to forget a system which has proved itself to be strong and reliable. Instead, we recommend GPC keeps their core culture, specifically the clan, alive as it transitions its business model. This will help to keep a consistent brand focused on a deep connection that feels like family.

That said, some change is critical. The company's current sales team is older, relationship-driven, and resistant to digital tools. Many sales employees have ignored online leads, which directly hurts PlantLuv's growth strategy that relies on digital sales. If this team does not evolve, PlantLuv will not be able to either. The culture doesn't need to change, but how it's expressed must. Relationships are still at the forefront, but they need to

grow to add CRM follow-ups, Zoom consults, and online communities. The definition of “personal connection” must stretch beyond in-person visits and fishing trips. Success here will result from staying rooted in Clan values while modernizing their delivery to higher technological standards.

GPC Critical Values

The first critical value of GPC we noted is GPC’s importance placed on customer relationships and service. Customer relationships have always been GPC’s foundation. However, as PlantLuv enters the market, customer service must be just as strong online as it is in person. Their traditional way of business needs a digital upgrade. Two actions we recommend for GPC to enact are a Service Relationship Policy and Punch Card Compensation Program. These two systems are explained in more depth below, and the values these two actions would promote.

Up to now, internet leads were placed in an inferior importance in comparison to in-person sales, and in some cases, online leads were ignored completely. This cannot be accepted as GPC transitions to their digitized product, the PlantLuv system. We recommend implementing a clear policy for service relations. This policy will prohibit any discrimination against online interactions. We recommend tracking this through CRM benchmarks, like time taken for online responses, conversion rates of online purchases, and their online customer satisfaction scores. These benchmarks can then be reviewed during the employees' annual performance review. If they succeed at meeting digital benchmarks, we recommend attaching a financial bonus to incentivize this change. The second system we recommend GPC enacts to aid customer service and relationships is a Punch Card Compensation Policy. This is a system where every time a sales employee finalizes a sale online, they can receive a “punch” on their card that denotes this sale. If the employee achieves 10 online sales per quarter, they can receive an additional bonus at the end of the year or PTO.

Through these two actions, we believe Customer Service & Relationships will flourish. Some specific behaviors we expect to occur are stopping of discrimination against online customers, faster availability for online interactions, GPC’s online communications will grow to feel more friendly and warm, and a higher state of accountability for digital business operations.

The second value we believe to be critical to GPC is their strive for innovation. The new product launch for PlantLuv is a shift for GPC to grow into a company that is known for their technologically driven product. As this becomes part of their persona, it is vital that GPC integrates innovation at each level of their business. This must not be an afterthought, but an intentional consideration. We recommend two ways to achieve this successful integration, by embracing change and providing additional training for employees in technology.

Through embracing change, this creates an environment where employees can feel safe to enter the unknown. GPC should celebrate learning and growth, creating a space where employees are not scared to be confused on a product or innovation, but rather excited to learn something new. While this is a simple change to the mindset and wording surrounding company innovation, it will adapt the way employees view tasks, making them seem less daunting, and thus easier to integrate into daily operations. It is important to note that for this to be successful, no employee may be scrutinized for a new idea failing. Rather, GPC must celebrate the attempt and innovative thinking. It is also recommended to foster this innovation, to provide more in-depth training and support systems for technology systems to employees. By doing so, this will give employees a resource to rely upon to grow their understanding and fall back on when confusion arises. Having a pre-established program to troubleshoot confusion will further eliminate fear associated with the unknown and innovation.

Through these two actions of embracing change and supporting innovation through education services, it is expected that behaviors to support innovation will arise. Some that we expect are a greater openness to change and technology, a lesser sense of fear for the unknown, a greater state of trust between employees and management, durability in changing environments, and a greater openness to new ideas/concepts/and people.

Job Descriptions and Analysis

Aligning Employment with the PlantLuv System

GPC has previously worked with a different market than their current target for the PlantLuv System. While some customer values do overlap, such as a shared interest in botanicals, the market's differences will require an alternate strategy from all aspects of the company. This connects to employees on the PlantLuv team to optimize compatibility between PlantLuv's customers and their business practices. Consequently, it is

recommended to hire a new sales force for the success of PlantLuv’s market integration. An important place to begin this hiring process is at the job description. A job description can aid applicants by guiding those most suitable to the job and give applicants a proper understanding of company culture, job expectations, and the competencies necessary for success in this role. For a PlantLuv Salesperson, we recommend focusing on applicants who have long-term experience in sales, have strong interpersonal skills, and strength in technological and botanical knowledge. A mock job description is listed below of how it is recommended to proceed with this process.

Job Description

Within this job description, it is recommended to emphasize the importance of a knowledge in botany, interpersonal skills, and a confident understanding of software as these are most vital to success in the role and consequently, the company’s financial success. A knowledge of botanicals will allow the sales team to provide educated insights to our clientele and explain the science behind the product. Interpersonal skills will equip our sales team to create genuine relationships by acting appropriately and kindly with our customers. A confident understanding of technology allows our sales team to advise our customers on any technology issues that may arise and with initial installation and maintenance processes.

Job Title	<i>PlantLuv Salesperson</i>
Company Overview	<i>Georgia Plant Company is an organization focused on providing high quality, thorough customer-service, and product support to ensure the customer receives the best experience. Started by Dr. Agarwal, a passionate botanist, nearly 30 years ago, Georgia Plant Company has been dedicated to the health of plants since our origin. First with the launch of our first-mover commercial spreader with high accuracy and efficiency, we have now moved to PlantLuv, a technology centered home gardening system that customizes the nutrients used in the fertilization to the plants and soil, enabling the health and wellbeing like never before. Focused on innovation, quality, and engagement, we promise to provide the best quality possible to our customers.</i>
Position Description	<i>With the launch of our new PlantLuv system, we need a team of salespeople to convey our message to our customers. These employees will need to be competent in their understanding of technology; particularly behind the application behind the product, be well-experienced in customer service and relations management, have a vast understanding of botanical knowledge, and the ability to perform simple maintenance and calibrations. With these skills, our sales team can help Georgia Plant Company successfully launch PlantLuv by educating the market on this new technology while ensuring proper quality is maintained.</i>
Key Responsibilities	Communicating product value to customer Advocating for added value from switching to our product Initial product setup Understand and explain to customer the type of plant and nutrients necessary for growth Troubleshooting technological issues Selling and finalizing the sale of the PlantLuv System
Knowledge, Skills, and Abilities	Botanical Knowledge Software Proficiency and Competency Strong Interpersonal Skills Ability to Collaborate Efficiently With Others Time Management Proficiency
Education and Experience	Bachelor's Degree (required) note: preferred major of Professional Sales or Biology Masters Degree (not required, but highly desirable) note: preferred major in Business/Technology/Biology 5+ Years in Customer Service (required)

Going further than these skills and expertise, it is also recommended to find applicants whose personalities match cohesively with the role and organization. These applicants are suggested to be extroverted, agreeable, and a high self-monitor. Firstly, extroversion is vital for success. A sales role is built on the foundation of communication. Through communication, sales teams can build an intimate bond between the customer and the company as they feel their interaction with the face of the company, our sales team, is genuine. This heightens trust and loyalty, leading to a reliable customer base. The sales employee must be easy to communicate with, engaging with the customer, and kind. Through extroversion, the customer can enjoy their conversation leading to a more positive association with our branding and company.

A second trait important for employees to have in sales is to be agreeable. Agreeableness will lead the sales member to be neutral to the customer's opinions. By doing this, it ensures they can meet the customer where they are at, conveying a conversation without any conflict. This will help establish strong relationships to increase loyalty. Additionally, people who are agreeable tend to have a high emotional competency and this control and understanding of emotions can guide these employees when maneuvering customer interactions.

Connecting off agreeableness, high self-monitoring can also prove to be successful in sales. This pattern of behavior clues the sales team into external cues in the conversation. With this information, someone with high self-monitoring would be able to react accordingly to become what the customer wants them to be. This trait would allow the salesperson to be able to customize their approach to every customer, allowing each person to receive a personalized interaction.

Two traits that are best to minimize or avoid are neuroticism and introversion. Those who are neurotic frequently feel intense emotions such as anger or sadness. They are more susceptible to emotional shifts from outside sources, making them potentially unstable for sales as they will constantly be exposed to human interaction. These conversations may not always end with a positive outcome, so emotional instability could become a large risk to the persona of the company if our salespeople do not remain calm and friendly.

Introversion is also a skill that should be avoided as it could harm the success of the salesperson. This role must thrive off communicating with people, and these customers are not typically familiar to the salesperson

prior to the role. Thus, if the employee is introverted, this could harm their success in meeting and speaking to new and returning customers.

Recruitment and Selection

Optimizing Employee Acquisition

It is necessary for an employee to be equipped with proper skills, abilities, and knowledge to effectively function within an organization. A proper selection system will guide the GPC hiring team to make the right choices as to who to add to their team.

When focusing on where and how to recruit, we recommend keeping PlantLuv's target audience at the forefront of all decisions as we need to find the subset of people who can best adhere to their needs. For PlantLuv, there are two target audiences: Gardening Elders and Garden Proud. Gardening Elders are retired, wealthier individuals who use gardening as a pastime and are very knowledgeable in botanicals. Garden Proud are also wealthier individuals who are still working, but their children are older and left the home. They are very focused on the aesthetics of their garden and this heightened interest makes them inclined to want to learn more about botany. Both groups' financial status gives them the flexibility to purchase a product like PlantLuv, and for this reason they expect personalized and expert service for the price-point they are paying. Our salespeople must be able to connect with this group, and to do so it is recommended to focus on salespeople with extensive years of experience and an in-depth knowledge of botanicals and technology. PlantLuv's market will be inquisitive as their level of care for the product is much higher than a typical gardener's. The salesperson must meet these questions with thoughtful, educated responses to seal the sale and communicate the value of the product to the consumer. Furthermore, to be a successful salesperson for the PlantLuv system, the employee must be more formal than GPC's previous salespeople. The PlantLuv's new market is not compatible with the previous sales force as their informal, male-dominated environment will not suite the market of Garden Elders and Garden Proud. Thus, it is recommended to recruit these employees in an in-person manner. An initial resume check can help with the bulk of sorting applicants; however, an in-person interaction can allow the hiring team to experience a genuine interaction where the applicants' manners and

formalities can be highlighted. This formal expectation for applicants can also be explained in the initial job posting through its tone. This will emphasize the value PlantLuv places on formalities to adhere to our target market, giving the applicants a better understanding of the ideal candidate.

It is recommended to ensure those hired are the best fit alongside the company's culture, job requirements, and other employees to find those with the highest probability of success. By keeping this at mind alongside the target consumer, this perspective can guide the process of recruiting to help GPC with how and where they will hire this new team.

Applicant Selection Process

When engaging in recruitment, we need to ensure that we are maximizing fit by hiring employees who have the skill, abilities, and understanding to perform sufficiently with the firm. This goes in three dimensions, the person to job, the person to group, and the person to organization. Specifically, this fit revolves around will the job motivate the person, will the person work efficiently with the group and will the applicant match the company culture. To maximize fit, GPC can utilize different types of tools such as screening, evaluation, and contingent.

The first step GPC can take for this recruitment process is to focus on the screening tools. After a job description is developed and posted to various sources, we will have the greatest number of applicants in our pool. To help whittle applicants to those most suitable for the job, screening tools are very efficient. For Georgia Plant Co., the first step of the selection process will be to pass an application scan. This will cost the company \$100 per applicant, but only provides criterion-related validity, or an accurate prediction of job success, of .35. However, this is a relatively inexpensive first step to rid of applicants who are an obvious irregularity in the company's preferred characteristics. This step will look for hard skills specifically, such as the applicant's prior experience, education, and interests, such as botany.

Those who pass the first test, will then move on to the structured interview. Here, the manager or hiring team can get to know the applicant in a more personalized manner. Noticing responses to questions such as their personal goals, where they see themselves in five years, or how they work with others can give insight to their personality and ambition, which can guide the interviewer to decide whether the applicant will fit with the company culture. This conversation can also be insightful to whether their interpersonal skills are up to par

with their future coworkers and customers. This step will cost GPC \$175 per applicant with a criterion-related validity of .51. While the cost is heightened, the validity or accuracy of predicting future success is nearly 50% justifying why this process is worthy of the price jump.

If an employee passes the structured interview, they may then move onto the final round of the selection process of a job sample test. This is a process where the employee will shadow a current employee and attempt to make their first sale. The current employee will take note of the applicant's success as the ultimate test of whether they have the personality and potential to be successful in this role. Things such as their interpersonal skills, professionalism, respect for the brand and respect given to the customer will be weighed to reach the conclusion of pass or fail of this step. If they pass, the applicant will then be transferred to a new hire and undergo a more in-depth training program where the company can provide tools for their success and personal growth. This test is the most expensive at \$300 per applicant and for this reason is only done for the most serious potential future employees. However, it is rated at a .64 criterion-related validity proving to be extremely predictive of future job success. This tool's accuracy makes it worthy of the cost.

The entire process will contain three steps, the application, structured interview, and job sample test. All of which will sum to a total cost of \$575 per hired employee. The process is designed to increase in cost, and risk to the company, as the applicant pool becomes more realistic of a successful hire.

Performance Management

Employee Success Through Performance Management

For the PlantLuv sales team's performance management method, it is recommended to use a combination of a relative approach with a forced distribution and a results approach. A relative approach where a portion of employees are organized into a fixed number of ranks, will allow GPC to have black and white succession planning and extensive administrative data. This is vital to have in abundance because as a product and team are launching, plans are subject to change with trial and error. This type of performance management system will highlight people who are naturally excelling, bringing attention to employees that could make great leaders and trainers in future operations. This can also be helpful to see patterns of what techniques are used with the more successful groups to recognize any patterns that can aid the product line's overall success. This ranking can also create competition between employees. If handled correctly, this can cultivate a culture of

extrinsically motivated employees who want to increase their success in sales, pushing the sales force to aim to continue growing as an employee.

This can specifically measure the sales team member's success out on the field. The ranking can be done based on a weighted average between the level of importance management places on a skill and the employees' results over the past quarter. These results could include their number of sales, amount of CRM benchmark successes, and ratings given from customers on dimensions such as the employee's knowledge of botanicals, technology, and the product, as well as their interpersonal and professionalism skills during the sale. This weighed average can then be used to place the employees into a ranking.

However, a relative method does not give a great deal of developmental feedback. This could leave employees frustrated with the lack of guidance from upper management, which may be further exacerbated by a competitive culture. For this reason, it is recommended to intertwine a secondary results approach where managers and employees define individualized goals and help and evaluate employees on their success in achieving said goals. This process can help employees obtain direction in their pursuit of growth by personalizing the steps to success. This can also allow upper management to see firsthand what frustrations their team is facing, giving the opportunity to troubleshoot and adapt processes to ensure strategic congruence is prioritized.

This addition of a results method can help fill the developmental gaps left behind by the relative method. During the goal setting meeting between employees and managers, the two parties can align their goals to find an area where both parties can achieve their expectations. This can also allow the manager to measure their own perceptions of the employee's success in their understanding of the product, skills in communicating, and overall proficiency and determination to better themselves as an asset to the company. The secondary measure can ensure competition and customer opinions do not overpower the employee's performance measurement and allow for a more holistic measurement with heightened reliability.

System Tradeoffs

When developing this performance management system, a trade-off occurred between acceptability & usability for a strengthened adherence to strategic congruence, validity, and reliability. When combining two methods and having two separate moving parts, the system became more complicated to use, lessening its usability.

However, it is still recommended to do so as it's benefits of employee growth from the combined system outweigh the loss of ease of use. For example, the second method's addition creates a special emphasis on strategic congruence where the parties can work together towards a goal. While this takes more effort, the financial outcomes of a specified direction will outweigh the loss of time and heightened effort. Furthermore, by combining management and consumer perspectives of employee performance, the system can cultivate a more accurate reflection of employee performance by ensuring the big picture can shine through. This process of multiple raters also ensures that one person cannot sway the measurement, heightening the reliability.

This process achieves strategic congruence through the goal-setting coordination between employee and manager. Together, these two can align their goals to the betterment of the company and ensure they stay on target for their overall strategy. This form of communication also prepares the system to naturally advise employees on their own personal development. The goal-setting process is a double-edged sword finding direction for both corporate and personalized initiatives. Also, the nature of the customer-based ranking system can also provide administrative data for hiring and firing of employees, and the results-based process can provide more detailed and individualized performance reports in terms of strategy. This detailed and eagle-eyed combination allows for the administrative data to meet a greater variety of needs depending on their level of depth.

Motivation and Change

Bonus-Incentives for Current Sales team

GPC is comprised of older employees with traditional values who emphasize customer relationships and interpersonal connections. The best way to align their goals of connection with our mission of bringing love and care into every household through our PlantLuv system is to showcase the immense connectivity and ability of the Internet.

We propose a bonus plan towards that goal known as the "Digital Pioneer Program" incentive program. This bonus plan is designed to slowly shift employee bias against Internet leads and provide immediate incentives that reward high-quality engagement, making internet-sourced sales financially more attractive to employees.

The “Digital Pioneer” Program is a multi-tiered bonus structure where payouts to the current employees are connected to their accomplishment in each step of the process to complete an online sale. All activities must be logged into the company CRM to be verified and checked for eligibility. Once checked, these bonuses can be either paid out immediately or added to the employee’s year-end bonus.

Len Agarwal’s father created GPC with a traditional, relationship-based sales model in mind, embodied by the older composition of the current sales force. If GPC decides to utilize its current sales force to sell the PlantLuv system, a significant organizational change will be necessary to shift from a traditional, relationship-based sales model to one that is more internet-oriented. Overcoming the current sales force's resistance will require a thoughtful and strategic approach.

Stakeholder Analysis & Change Mangement

A high-level stakeholder analysis reveals the different groups that will be impacted by this change and their current disposition. A graphic depicting the stakeholder analysis and each group's stance is listed below. To move the current sales force and their managers from detractors to promoters, we need to implement a multi-faceted change management strategy.

Stakeholder Group	Current Stance	Rationale	Target Stance
Current Salesforce	Detractor	The salesforce is comprised of an all-male, older demographic (youngest is 42) who value personal relationships with clients, exemplified by Sales Manager Bob's "up close and personal" approach. They are accustomed to a slower-paced business and are suspicious of internet leads, to the point of ignoring them. Their established routines, like gathering at a co-owned cabin, indicate a tight-knit culture resistant to new ways of working.	Promoter
Sales Manager (Bob)	Detractor	As the leader of the current sales team, Bob has been instrumental in hiring and training the entire force in his "up close and personal" style. Rumors suggest he has explicitly told his team to ignore internet leads, and though he denies it, his actions and the team's behavior indicate a clear bias against an internet-based sales approach.	Promoter
Internal Sales & Marketing Team	Promoter	This team is frustrated with the current salesforce's refusal to follow up on internet-generated leads. They are already engaged with the company's website and are forwarding leads to wholesalers and distributors to ensure potential customers are contacted. They would benefit from a salesforce that embraces online channels.	Promoter
Director of Human Resources (Josh Lee)	Neutral to Promoter	Josh Lee is concerned about the sales force's culture and their ability to adapt to a new, tech-focused product and a different customer base. He recognizes the need for a different approach for the PlantLuv System and is exploring creating a new salesperson position, indicating he is open to and likely supportive of the necessary changes.	Promoter
Top Management (Len Agarwal, Amy Bowes)	Promoter	Len Agarwal is driving the launch of the tech-driven PlantLuv System to penetrate new markets. Amy Bowes, the CMO, is considering online ordering and other digital strategies, showing a clear commitment to an internet-oriented approach for the new product.	Promoter

When implementing this new product, it is vital to have all of GPC's employees as advocates for the change.

Otherwise, this can cause conflict for the organization. To prohibit this, we recommend focusing on a five strategy-process to stop any unnecessary friction.

1. Education and Communication:

- Rationale:** The sales force's resistance stems from a fear of the unknown and a belief that their current methods are superior. Clear communication can address these fears and foster understanding.

- **Actions/Strategies:**

- o **Leadership Alignment:** Len Agarwal and Amy Bowes need to clearly and consistently articulate the strategic importance of the PlantLuv System and the necessity of an internet-oriented approach for its success.
- o **Vision Casting:** Create a compelling vision for the future that shows the sales force how their roles will evolve, not become obsolete. Emphasize that their deep product knowledge and relationship-building skills are still valuable but need to be applied in new ways.
- o **Transparent Communication:** Hold town hall meetings and workshops to explain the new system, the target market, and the rationale for the new sales process. Address their concerns directly and honestly. Josh Lee, with his strong knowledge of company and employee relations, should play a key role in these communications.

2. Participation and Involvement:

- **Rationale:** Involving the sales force in the change process can increase their sense of ownership and reduce resistance.
- **Actions/Strategies:**
 - o **Appoint Change Champions:** Identify one or two of the more open-minded salespeople from the existing team to be part of a pilot program for the PlantLuv System. Their success and positive experiences can influence their peers.
 - o **Seek Input:** Involve the sales team in developing the new sales process. For example, ask for their input on how to best integrate online leads with personal follow-up to maintain a "personal touch."
 - o **Leverage Bob's Influence:** While Bob is a detractor, he is also a respected leader. The leadership team should work closely with him to understand his concerns and involve him in shaping the new strategy. His conversion to a promoter would be highly influential for the rest of the team.

3. Facilitation and Support:

- **Rationale:** The sales force will need new skills and resources to succeed in an internet-oriented environment. Providing adequate support can build their confidence and competence.
- **Actions/Strategies:**
 - **Comprehensive Training:** Provide extensive training on the PlantLuv System's technology, the proprietary website, and the new CRM system Amy Bowes is considering. Training should also cover digital communication skills and how to effectively engage with customers online.
 - **Resource Allocation:** Equip the sales team with the necessary technology, such as laptops or tablets, and ensure they have reliable internet access.
 - **Dedicated Support Team:** Create a dedicated technical support team that the sales force can easily contact for assistance with the new systems.

4. Negotiation and Agreement:

- **Rationale:** To motivate the sales force to adopt new behaviors, their compensation and incentives must be aligned with the new goals.
- **Actions/Strategies:**
 - **Revised Compensation Structure:** Implement the bonus plan designed to encourage following up with internet-generated leads. Furthermore, adjust the overall commission structure to reward activities related to online sales and engagement, not just large orders. This directly addresses their potential disgruntlement over web sales.
 - **Performance Metrics:** Introduce new performance metrics that reflect the desired behaviors, such as response time to online inquiries, customer satisfaction with online interactions, and conversion rates of internet leads.

5. Coercion (as a last resort):

- **Rationale:** If significant resistance persists after implementing the above strategies, it may be necessary to make it clear that adoption of the new methods is not optional.
- **Actions/Strategies:**

- **Clear Expectations:** Management must clearly state that following up on internet leads and utilizing the new systems are mandatory parts of the job.
- **Performance Management:** The newly designed performance management system for PlantLuv salespeople should be used to hold them accountable for their performance against the new metrics.
- **Personnel Changes:** If certain individuals, including leadership like Bob, continue to actively resist and undermine the new strategy, Josh Lee and top management may need to consider reassigning them or, in the worst-case scenario, letting them go.

By implementing these integrated strategies, GPC can effectively manage their transition to PlantLuv and increase the likelihood of their current sales force embracing this new methodology.

Behavioral Structured Interview Question Recommendations

Reaching back to our recommendation for approaching the selection process, we urged GPC to use a structured interview as one of the hiring steps. Below there are recommended questions, highlighting the competency they measure, and a 5-point rating scale to score applicant answers.

The first recommended question is as follows: “Tell me about a time when you had to explain a complex scientific or technical concept to someone who was not an expert. How did you make sure they understood the information, and what was the outcome of the conversation?”

This question measures the candidate’s ability to not only understand but translate their Botanical and Technical Knowledge into simple and persuasive language for a novice home gardener. It also measures effective communication and whether they are attuned with GPC’s customer-centric mindset.

5-Point Rating Scale:

- **5 (Excellent):** The candidate clearly describes the situation using the STAR method (Situation, Task, Action, Result). Their *action* demonstrates a deliberate strategy, such as using analogies, avoiding jargon, asking clarifying questions to check for understanding, and focusing on the benefits relevant to the customer. The *result* is a customer who feels confident and empowered, leading to a positive outcome (e.g., a sale, increased trust, or a resolved issue).

- **4 (Good):** The candidate provides a solid example and explains the concept effectively. They may be slightly less detailed about their specific strategy for ensuring comprehension but clearly achieved a positive result.
- **3 (Acceptable):** The candidate provides a relevant example, but the explanation of their method is generic. They explain what they did, but not necessarily *why* they chose that approach. The outcome was adequate, but not exceptional.
- **2 (Marginal):** The candidate struggles to provide a clear example or describes a situation where their explanation was only partially successful or left the customer confused. They may have focused more on demonstrating their own expertise than on ensuring the customer's understanding.
- **1 (Unacceptable):** The candidate cannot provide a relevant example, or the example shows they failed to communicate effectively, became frustrated, or created a negative customer experience.

The second recommended question is, "Describe a situation where you had to help a frustrated customer troubleshoot an issue with a piece of software or technology. What steps did you take to diagnose the problem, and how did you guide them toward a resolution while managing their frustration?" This question evaluates the candidate's Software Proficiency in a real-world application, their problem-solving skills, and their ability to perform simple maintenance and calibrations under pressure. It also assesses their dedication to customer service and relationship management.

5-Point Rating Scale:

- **5 (Excellent):** The candidate details a logical, systematic approach to troubleshooting. They demonstrate empathy for the customer's frustration, actively listen, ask clarifying questions before jumping to conclusions, and patiently walk the customer through the steps. They confirm the resolution and ensure the customer is satisfied. Their focus is on both solving the problem and repairing the customer relationship.
- **4 (Good):** The candidate describes successfully solving the technical issue and calming the customer. Their troubleshooting process might be less structured, but they still demonstrate patience and achieve a positive result.

- **3 (Acceptable):** The candidate provides an example of solving a technical problem but shows little focus on managing the customer's emotional state. They got the job done but may have missed an opportunity to build a stronger relationship.
- **2 (Marginal):** The candidate's example shows they struggled with the technical aspects or became impatient with the customer. They may have had to escalate a simple issue or left the customer feeling unintelligent.
- **1 (Unacceptable):** The candidate cannot provide a relevant example or describes a situation where they blamed the user, showed obvious frustration, or failed to resolve the issue, damaging the customer relationship.

A final example for the structured interview questions is “Tell me about a time you faced a skeptical or dissatisfied customer who was challenging your product's value. How did you adapt your communication style to address their concerns, and what was the result for the long-term relationship, even if you didn't make an immediate sale?” This question assesses the candidate's Agreeableness and High Self-Monitoring by probing their ability to handle conflict and adapt their approach based on external cues. It directly measures their dedication to establishing and maintaining relationships with customers, even in difficult circumstances.

5-Point Rating Scale:

- **5 (Excellent):** The candidate describes staying calm, professional, and non-defensive. They actively listened to and validated the customer's concerns ("I understand why you feel that way..."). They clearly explain how they adapted their approach—perhaps shifting from price to long-term value, or from data to a personal testimonial. The *result* is a de-escalated situation where the customer feels heard and the relationship is preserved or even strengthened, regardless of the immediate sales outcome.
- **4 (Good):** The candidate provides a strong example of handling a difficult customer effectively. They may be less explicit about *how* they adapted their style but clearly managed the situation professionally and maintained a positive rapport.
- **3 (Acceptable):** The candidate describes a situation where they handled the objection but in a more formulaic or rigid way. They didn't make the situation worse, but they didn't necessarily turn it into a

positive relationship-building moment. The focus was on countering the objection rather than understanding the customer's perspective.

- **2 (Marginal):** The candidate's example shows they became somewhat defensive or entered into a debate with the customer. They may have "won" the argument but damaged the relationship in the process.
- **1 (Unacceptable):** The candidate cannot provide an example or describes getting into an argument, dismissing the customer's concerns, or becoming visibly agitated. They demonstrate an inability to separate their personal feelings from a professional interaction, indicating low self-monitoring and agreeableness.

Marketing Analysis: GPC and the PlantLuv Product

Evaluating STP

Target Segmentation

To launch PlantLuv in the consumer gardening market, GPC must ensure the product is targeted toward customer segments that are both open to adopting new technologies and capable of driving meaningful market penetration. To identify the most promising target audiences, we recommend a combination of demographic segmentation, specifically the income level and garden size, and psychographic segmentation, including interest in gardening, expertise, and lifestyle. This segmentation method leads to the recommendation of targeting the Gardening Elders and Garden Proud segments for both initial market entry and long-term growth.

Gardening Elders: Gardening Elders are experienced gardeners who own large gardens and possess high level of wealth and expertise. For them, gardening is a meaningful hobby that they refine over time. They tend to grow a variety of different plant types and species, demonstrating the sophistication of their gardens. As the third wealthiest segment, and part of the Plant-Expert, High Enjoyment behavioral profile, they account for 18% of garden retail spending. The need for achieving garden sophistication combined with their high purchasing power and motivation, makes Gardening Elders an excellent fit for convenient yet personalized, data-driven solutions like PlantLuv. Additionally, Gardening Elders enjoy sharing their expertise with others, making them likely to engage actively in PlantLuv's online forum and community features. Their credibility, commitment, and influence position them as ideal innovators and early adopters, essential for helping GPC enter and establish itself in the consumer gardening market.

Garden Proud: Garden Proud consumers are wealthy homeowners with large gardens that feature a wide variety of plants and design elements. As the wealthiest segment responsible for 28% of total garden retail spending, the highest of any group, they have the financial flexibility to explore new innovations, making them well-positioned to adopt premium products like PlantLuv. Although this group is categorized as Plant-Novice, High Enjoyment individuals, some have developed expertise over time as their interest in garden maintenance deepens. Influenced by trusted influencers, such as Gardening Elders or experienced peers within their

community, Garden Proud consumers are a strategic segment for “crossing the chasm” into the early majority. Their growing interest in gardening makes them likely to seek tips, actively engage in the PlantLuv forum, and make full use of the system’s video tutorials, care reminders, and expert content. Their combination of motivation, high spending power, and aspiration makes them a key segment for expanding market reach beyond early adopters.

Personas

Jessica Smith

Gardening Elder



Brands



Background

Age	57
Education	High school graduate
Gender	Female
Ethnicity	Caucasian
Marital Status	Married
Occupation	Retired real estate agent
Income	High
Living Situation	Large home with backyard garden
Location	Bloomington, Indiana

Cautious but curious

Influential

Patient

Disciplined

Vals: Thinker

Interests

- Watching and reading gardening programs and magazines
- Growing vegetables, manicuring shrubs, and experimenting with new plant varieties
- Attending local gardening clubs and community events

Pain Points

- Wants to make data-backed decisions about soil and plant care, but doesn’t fully trust online tips or generic advice
- Finds it tedious to manually manage all gardening information
- Aging presents physical and cognitive challenges

Goals and Motivations

- Aims to expand gardening knowledge without relying on unverified tips
- Looks for tools and products that simplify or enhance routine care

Rachel Green Garden Proud

Brands



Vals: Achiever

Interests

- Enjoys weekend yard work but hires help for major landscaping tasks
- Cares deeply about curb appeal and aesthetics, often planning seasonal layouts
- Often finds inspiration on Instagram and Pinterest

Pain Points

- Has limited expertise in gardening
- Finds it difficult to choose the right fertilizers or care plans for her plants
- Frustrated when plants underperform despite spending on high-end products

Goals and Motivations

- Aims to expand gardening knowledge without spending too much time or effort searching for it
- Wants her garden to look as nice as those seen on social media

Background

Age	43
Education	High school graduate
Gender	Female
Ethnicity	Caucasian
Marital Status	Married
Occupation	Cupcake shop owner
Income	High
Living Situation	Large home with backyard garden
Location	Austin, Texas

Creative

Expressive

Prideful

Detail-oriented

Value Proposition Statement

To Gardening Elders, PlantLuv is the personalized plant nutrient delivery subscription that allows them to confidently refine their expertise and improve their garden using scientific data-driven insights.

To Garden Proud gardeners, PlantLuv is a reliable system that delivers personalized care tips, expert video guides, and tailored recommendations that helps them grow vibrant, beautiful gardens with confidence, even without extensive prior experience.

Customer Profitability

Segment Profitability

Segment Size: Assume that there are 130 million households. When 55% of American households engages in gardening activity, and Gardening Elders and Garden Proud takes up a proportion of 14% and 17% respectively, Gardening Elders' segment size is 10.01 million and Garden Proud size is 12.16 million.

Adoption Rate: Based on weekly hours spent gardening (12+ hours), we estimate an adoption rate of 45% for Gardening Elders and 58% for Garden Proud.

Purchase Behavior: At \$35 per unit, Gardening Elders are expected to purchase 3 units/year, once every season except winter. Garden Proud will purchase 4 units/year as their image-conscious habits may make them reapply the nutrient spikes more than once during the seasons.

Profit Margin: Taking an assumption of 10 years, according to the finance department, the average unit margin is 8.63 (\$35 sales price – (average of total variable cost/10)). Profit Margin is estimated at 24.67%.

Fixed Cost: According to the finance department, the average total fixed cost throughout 10 years is \$315,000. Allocated by segment revenue share (32.4% vs. 67.6%), Gardening Elders are assigned \$102,048, and Garden Proud \$212,952 of fixed costs for the first year, totaling \$315,000.

Profitability:

	Gardening Elders	Garden Proud
Segment Size	10,010,000	12,155,000
Segment adoption %	45%	58%
Purchase behavior		
Price	35	35
Annual frequency	3	4
Profit Margin %	24.67%	24.67%
Fixed Costs	\$102,048	\$212,952

Profitability	\$116,580,267	\$243,276,494
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By multiplying the segment size, adoption rate, purchase behavior, profit margin rate and deduct fixed cost, we achieve a segment profitability of \$117 million for Gardening Elders, and \$243 million for Gaarden Proud.

Increasing Segment Profit

Gardening Elders

Adoption rate can also be increased by reinforcing this segment's self-image as gardening experts. Gardening Elders take pride in their knowledge and experience, and they value tools that align with their skills and knowledge. Positioning PlantLuv as a scientific support system that validates their expertise makes them feel like the system is a natural extension of their capabilities. When they see PlantLuv as a product that respects and reflects their gardening identity, they are more likely to adopt it, directly increasing the segment profitability.

Garden Proud

Increasing adoption is the most practical and effective way to grow profitability within this segment. This segment places a high value on garden aesthetics and is willing to spend to maintain a visually impressive garden. PlantLuv can appeal to them by showing how its tailored nutrient spikes help plants thrive and look their best in every season. By highlighting the simplicity of the process along with visual outcomes these consumers care about, including vibrant blooms, healthy leaves, and Instagram-worthy gardens, adoption rate can be raised.

Adding Value

Gardening Elders

Value can be added by rewarding badges to users on online forums. Gardening Elders are the most experienced segment and often see themselves as experts and influencers in gardening. They take pride in their knowledge and enjoy sharing tips, teaching others, and reinforcing their role as expert gardeners. When exclusive "Expert Contributor" badges are rewarded in the online community forum, extra value can be created as it appeals directly to their desire for recognition. These badges could reinforce their identity and

status within the gardening community, resulting in higher psychological and emotional value, increasing engagement and loyalty.

To further create value for Gardening Elders, PlantLuv can offer free, personalized customer support throughout the soil testing, nutrient selection, and subscription setup process. Gardening Elders tend to be less comfortable with digital platforms or subscription services. They value reliability, clear guidance, and hands-on service. Providing such customer service in their augmented product strategy not only improves customer satisfaction but also builds loyalty, enhancing perceived value to these tech-wary yet committed gardeners.

Garden Proud

To add value, PlantLuv should implement “before and after” photo comparisons on online forums and social media. As Garden Proud have high income levels and a strong interest in garden aesthetics, they are often motivated by self-actualization and sense of accomplishment through their garden's appearance. PlantLuv can create added value for this segment by emphasizing how it can help plants look their best throughout different seasons by providing tailored, scientifically backed nutrient care. When PlantLuv reinforces the feeling that they can achieve expert-level results using their product, it will meet their psychological need for beauty, pride, and mastery, increasing PlantLuv's perceived value.

PlantLuv should also use premium packaging made from recyclable and biodegradable materials to increase value to the Garden Proud segment. These consumers care about their self-image. They not only prioritize a visually appealing garden but also tend to see themselves as environmentally responsible and social reformers. By using sustainable packaging with elegant design, PlantLuv allows them to reinforce their identity as both stylish gardeners and nature protectors. This aligns with their values and adds pride, enhancing the product's value.

Pricing Strategy

Pricing Orientation

Given the current competitive marketplace, we recommend a customer-oriented premium pricing strategy, a strategy that focuses on meeting customers' needs and expectations.

The home gardening market has been experiencing significant growth but remains competitive. The COVID pandemic has created 18.3 million new gardeners, and 42% of gardeners spent more time gardening. Currently, there are two large national firms, Royal and Carebest, which collectively hold 75% of the market. Being one of the smaller companies accounting for 25% of sales in the market, the limited funds make it challenging for GPC to use low prices to compete against Royal and Carebest. Not only do stockholders' expectations on growth and profit make this idea even more difficult, but the lower price might cause skepticism among consumers as they believe cheaper solutions are not as reliable.

A customer-oriented pricing strategy is the most strategic approach to infiltrate this player-dominated market. Premium pricing creates an assumption of reliability and expertise, which is what our targeted segments value. When our targeted segments spend over \$2,000 annually on gardening-related products and services, a higher price will be acceptable if value is effectively conveyed. The customer-oriented approach allows GPC to meet customer's wants and needs while positioning itself as a premium solution in the growing gardening market.

Value-based Table

Incremental Benefits	Improved Value	Benefit Weight	Weighted Factor
Plant identification accuracy	20%	0.15	3%
Care reminders	15%	0.18	2.7%
Plant database	35%	0.09	3.15%
Health diagnosis	46%	0.14	6.44%
Growth tracking	10%	0.05	0.5%
Intuitive interface	28%	0.04	1.12%

Offline access	100%	0.05	5%
Soil measurement and diagnosis	100%	0.1	10%
Subscription of plant spikes	100%	0.2	20%
Overall		1	51.95%

Based on the table above, it is suggested that PlantLuv offers an overall improved value of 51.95% compared to its competitors. This improvement in value suggests that the price can be priced 51.95% above the competitor's price. However, GPC should understand that if its actual value is below perceived value, expectation disconfirmation might occur, potentially resulting in customer dissatisfaction and low repeated purchases.

Value-Based Pricing Approach

We recommend using a value-based pricing approach for PlantLuv, an approach that focuses on the product's perceived value. Unlike competitors, PlantLuv provides offline access, personalized soil measurement and diagnosis, and a subscription model for targeted nutrient spikes. These unique benefits, along with other incremental benefits shown in the value-based table above, result in a perceived value of 51.95% higher than competitors.

As both target segments have strong purchasing power and place emphasis on results and expert-backed solutions, they are not as price-sensitive or deal-driven. Instead, they are willing to pay a premium if the product delivers on its perceived value. As PlantLuv's value proposition aligns with both targeted segment's needs, perceived value will be high. When PlantLuv offers unique benefits, the perceived value will be higher than other competitors. Combined with the larger perceived value from premium pricing, our targeted segments will be willing to pay at a higher price, allowing the company to enjoy a larger profit margin.

Elasticity of Demand

PlantLuv's demand elasticity further strengthens the argument of using value-based pricing. According to the income-based approach, consumers tend to spend more on premium products as their income increases. Both Gardening Elders and Garden Proud are high-income segments with strong interest in gardening, making them more receptive to high-value, premium solutions like PlantLuv. Combined with the incremental benefits and higher perceived value PlantLuv offers, customer price sensitivity is low, and demand is relatively inelastic.

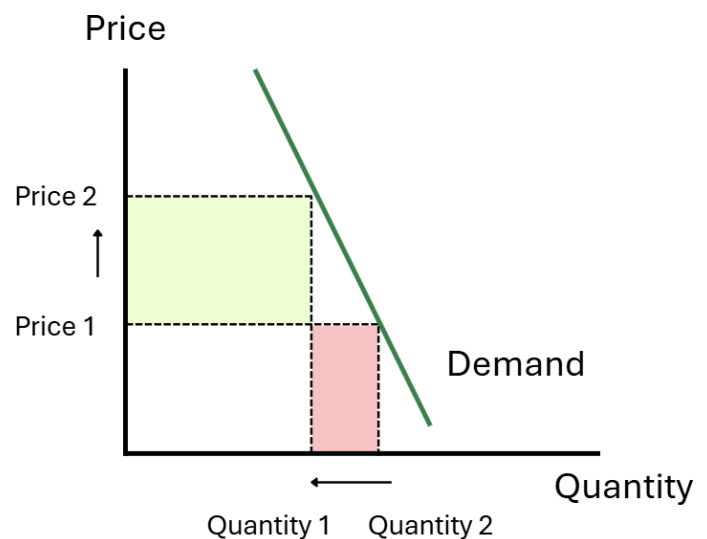
As shown in PlantLuv's inelastic demand graph (Table M1), when the price increases from Price 1 to Price 2, the resulting quantity decrease from Quantity 1 to Quantity 2 is proportionally smaller.

The green area represents the additional revenue gained due to price increase while the red area represents revenue lost due to reduced quantity.

As the gain from the price increase outweighs the loss from lower quantity, total revenue increases.

The inelastic demand supports the value-based pricing approach by ensuring that higher pricing will not impact buyers' purchase decision dramatically within the target segments. By pricing at its value or at a premium, GPC can capture more of the product's perceived value without significantly impacting transactions and increase revenue.

Table M1



Channel Structures

Direct VS. Indirect Channel Structure

Selling through PlantLuv's own sales representatives or directly to consumers online represents a direct channel structure, with no intermediaries involved. This approach allows PlantLuv to retain full revenue, leading to higher profit margins. It also provides direct contact with customers, enabling the company to gather feedback, tailor marketing strategies, and build stronger customer relationships. Selling through PlantLuv's

own sales representatives further allows shelf placement and sales messaging to be optimized. However, this model requires significant initial investment and time to establish. It also limits the distribution breadth, making it more difficult to be accessible and reach a wide customer base.

Selling through garden product distributors represents an indirect channel, where intermediaries handle parts of the sales and distribution process. This allows faster market entry and a wider distribution breadth as PlantLuv can leverage established distributor and retailer networks to access target markets more efficiently. Moreover, this also allows specialization, where the company focuses on product development while intermediaries manage sales, logistics, and customer interaction. GPC can also avoid costs associated with learning and acquiring new systems. Yet, this structure requires revenue sharing with intermediaries and less control over the customer experience. They have limited control over shelf placement, sales messaging, and how the product is presented to customers.

PlantLuv's Channel Structure

Given its small size, limited resources, and lack of prior retail experience, GPC should adopt an indirect channel structure for its initial launch. The indirect channel structure's lower initial investment and ability to reach customers nationwide allows GPC to build trust by leveraging the intermediary's brand image, enjoy a faster market entry, all at a lower cost.

Contractual Coordination

As a relatively small company with cautious shareholders who expect steady earnings growth, contractual coordination is a practical and cost-efficient strategy. It enables faster market entry with lower upfront investment and reduced risk. Through contracts and strategic partnerships with established retailers or distributors, PlantLuv can maintain control over its vertical channel relationships without the capital intensity of full ownership. This approach aligns well with the company's current resource constraints and growth expectations, allowing it to penetrate the market efficiently while keeping risk manageable.

Specialty Stores

GPC should launch PlantLuv through gardening specialty stores. PlantLuv is a system tailored to serious gardeners who are passionate and financially capable of investing in personalized solutions to enhance their gardens. These consumers are likely to seek expert advice on new gardening technology and invest in premium gardening solutions. These tend to be found in specialty stores which offer deep and narrow assortments and knowledgeable staff. When customers already associate these specialty stores with high-quality gardening solutions, it becomes more likely for them to discover and find PlantLuv. This strategic alignment between channel and customer expectation enhances visibility and builds trust during the launch phase.

Promotion

Gardening Elders

GPC should use an offline and passive IMC strategy to market PlantLuv to the Gardening Elders segment. This segment consists of older consumers typically less digitally engaged and more accustomed to traditional media like television and magazines. Offline, passive channels are more familiar to them and allow messaging to be delivered in a non-intrusive, comfortable, and trustworthy manner.

Public Relations and Direct Marketing

Specifically, Public Relations can be an effective tool to reach and communicate with this group. Gardening Elders are frequent viewers and readers of gardening programs, which they trust and rely on for information. By featuring PlantLuv on trusted media outlets, such as on BBC Gardeners' World, GPC can leverage the credibility and longstanding trust these platforms have established with their audiences. This exposure presents PlantLuv not as a sales pitch, but as a recommended solution from a credible source.

Direct marketing can also be an effective tool to communicate with this group. Gardening Elders already enjoy reading gardening magazines, so delivering a high-quality, informative brochure to their homes aligns with their existing habits. This method allows them to review the scientific and personalized benefits of PlantLuv at their own pace. By reinforcing memory through tangible materials, PlantLuv is more likely to enter their retrieval set and even their evoked set when they make purchase decisions.

Garden Proud

GPC should use an active and online strategy to communicate with the Garden Proud segment. Many Garden Proud consumers are “empty-nesters”, and they actively use digital channels, especially Facebook and Instagram, to stay connected with family and to showcase their gardens as a hobby and source of pride. An interactive and visually engaging strategy is best suited to this segment, as they are already comfortable engaging with brands through digital content and communities.

Online Marketing and Direct Marketing

Online marketing is one of the most appropriate IMC elements for reaching and communicating with the Garden Proud segment. By leveraging their habit of finding design inspirations online, publishing the reviews and comments of trustworthy home and garden influencers on blogs and social media will communicate to them in a natural and non-aggressive way, appearing more authentic, engaging, and credible to this segment.

Direct marketing, especially through mobile marketing, is also an effective tool to communicate with this segment. Garden Proud consumers are relatively more digitally engaged and responsive to messages.

Leveraging mobile marketing by sending short and personalized messages such as care reminders to their phones can reach them easily. These personalized nudges not only reinforce daily engagement but also increase the likelihood that PlantLuv enters their consideration during future purchasing decisions.

Measuring Traditional Media

We use gross rating points to measure when and how often Gardening Elders have been exposed to marketing communications, by multiplying the frequency and reach of exposure. For public relations, frequency can be measured by leveraging viewership data from programs PlantLuv featured in, including average number of viewers and the number of times it is aired. To measure reach, audience demographics can be measured. By using the age, income level, and address, the percentage of targeted segment that was reached can be estimated. For direct marketing, frequency can be measured by tracking the number of brochures sent and how often each household receives one. Reach can be determined by noting the total number of households the brochures are delivered to.

Measuring Web-Based Media

To measure the effectiveness of online marketing targeting the Garden Proud segment, web and social media analytics tools such as Meta Business Suite or Google Analytics can be used. By tracking the numbers of user views and their demographics, we can assess how effectively the Garden Proud segment are communicated to. For mobile marketing in direct marketing, effectiveness can be measured by analyzing the open rates, click-through rates, and conversion rates. For example, the company can track the percentage of new subscribers that has clicked on personalized care reminders. These insights can help determine the types of messages that are most effective in communicating, allowing the company to refine future promotions.

Social Media

Social media is a crucial tool for communicating with both of our targeted segments. Social media is effective to reach Gardening Elders as they often use it to share expertise and stay informed. Garden Proud also uses social media to get new aspirations to maintain and beautify their gardens. Using applications such as Instagram, Facebook, and Pinterest, to reach out to both segments can be effective, increasing sales and adoption rate.

For the Gardening Elders, GPC can leverage social media to collaborate with and obtain reviews from trusted experts, such as the host from BBC Gardeners' Word. When Gardening Elders see their favorite influencers appreciating and using PlantLuv, they are not only more likely to try it but might even start advocating for it.

For the Garden Proud segment, social media should feature aesthetic garden demos, including before-and-after garden transformation videos and seasonal design mockups. As the group who prioritize garden appearances, when these demos showcase their dream garden, they will tend to try purchasing PlantLuv to achieve similar results.

Operational Strategy: Driving Efficiency, Quality, & Growth

Capacity Management

Short-Term Tradeoffs vs. Long-Term Efficiency

To determine the most strategic production system for PlantLuv, we conducted a sensitivity analysis using the three projected Year 1 demand scenarios: pessimistic (50,000 units), baseline (100,000 units), and optimistic (175,000 units), along with a weighted average calculated using their respective probabilities, all of which are shown in the table below. In the baseline scenario, which aligns with our financial projections, the worker-paced system is less expensive during the first two years. However, with demand expected to grow by ten percent annually, reaching around 121,000 units by Year 3, the machine-paced system becomes the lower-cost option from that point forward. This scenario highlights the short-term tradeoff in cost versus the long-term benefit of investing in a scalable and efficient process.

Units	Machine Paced	Worker Paced	Cheaper Option	Difference in Cost
100,000	\$962,000.00	\$908,000.00	Worker	\$54,000.00
50,000	\$982,045.07	\$981,021.33	Worker	\$1,023.74
175,000	\$1,046,000.00	\$1,214,000.00	Machine	\$168,000.00
121,250	\$985,800.00	\$994,700.00	Machine	\$8,900.00

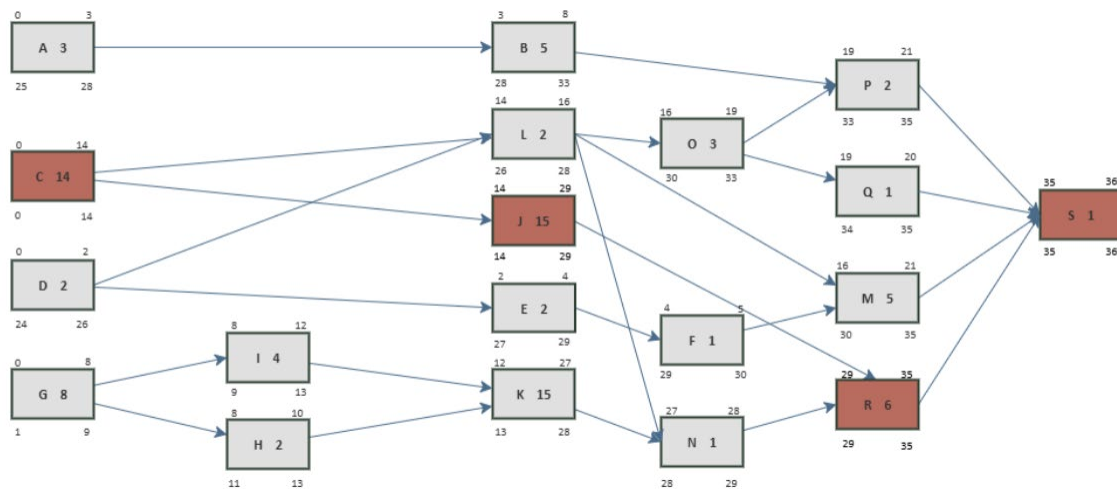
Cost Structure Analysis and Final Recommendation

The cost differences between the systems are primarily driven by their cost structures. The machine-paced line has a higher fixed cost of \$750,000 compared to \$400,000 for the worker-paced system, but its variable cost per unit is significantly lower (\$1.12 versus \$4.08). Both systems share the same compliance fee, so this difference in per-unit cost plays a major role in driving long-term savings. When we analyzed the weighted average demand across all three scenarios (121,250 units), the machine-paced option was already the cheaper choice in Year 1. Even in the pessimistic scenario, where the worker-paced system is initially cheaper by a wide margin, its advantage shrinks over time as demand grows. In contrast, the optimistic scenario strongly favors the machine-paced system from the start. Together, these results show that while the worker-paced line

may seem more economical under certain short-term assumptions, the machine-paced process offers better long-term value. Given Georgia Plant Care's expected growth and strategic goals, we recommend investing in the machine-paced production system.

Project Management

Network Diagram & Critical Path Identification



To prepare for full-scale production of the new project, we developed a network diagram (shown above) mapping all project activities and their immediate precedence relationships. Using the optimistic, most likely, and pessimistic durations from the project data, we applied the PERT formula to calculate the expected duration of each activity. This enabled us to identify that the **initial project duration was 36 weeks, with a critical path of C → J → R → S.**

Project Crashing Steps

Because this exceeds our targeted 29-week launch date, we implemented project crashing to reduce the timeline by 7 weeks. Our approach prioritized crashing activities on the critical path with the lowest cost per week, without going below the specified minimum durations. Here is a breakdown of the crashing process:

1. Crash J by 1 week to reduce the project from 36 to 35 weeks (\$500).

2. At this point, a second critical path — G → I → K → N → R → S — also emerged, both paths now sitting at 35 weeks.
3. Since R is shared across both critical paths, we crashed R twice (\$1,650 total), bringing it from 6 to its minimum duration of 4 weeks. This reduced the total project to 33 weeks.
4. Next, we simultaneously crashed J and G by 1 week each (\$1,300 total), bringing J to 13 weeks and G to 7 weeks, reducing the total to 32 weeks.
5. We repeated this in the next step, crashing J to its minimum (12) and G to 6, taking the project to 31 weeks.
6. At this point, both paths still had zero slack, so we crashed C and G by 1 week each (\$1,700 total), bringing C to 13 and G to 5, reaching 30 weeks.
7. Finally, we crashed C and G once more, taking C to its minimum of 12 weeks and G to 4 weeks, reducing the overall project duration to our target of 29 weeks.

Final Project Schedule

This structured crashing process ensured we minimized cost while maintaining feasibility. The total crash cost incurred was \$8,150. After crashing, we recalculated all early start (ES), early finish (EF), late start (LS), late finish (LF), and slack values for each activity. The updated schedule (shown to the right) confirmed that S ends at week 29 with zero slack, validating our adjustments. The blue cells represent critical path activities. This comprehensive approach ensures the project is on track for a timely and cost-efficient product launch.

Fully Crashed Project					
Activity	ES	EF	LS	LF	Slack
A	0	3	18	21	18
B	3	8	21	26	18
C	0	12	0	12	0
D	0	2	18	20	18
E	2	4	20	22	18
F	4	5	22	23	18
G	0	4	0	4	0
H	4	6	6	8	2
I	4	8	4	8	0
J	12	24	12	24	0
K	8	23	8	23	0
L	12	14	21	23	9
M	14	19	23	28	9
N	23	24	23	24	0
O	14	17	23	26	9
P	17	19	26	28	9
Q	17	18	27	28	10
R	24	28	24	28	0
S	28	29	28	29	0

Order Policy

EOQ, Safety Stock, & EOQ/ Total Inventory Cost Comparison

To evaluate the ordering policies for GPC's proposed circuit board suppliers, M2 and P3, we analyzed key inventory metrics using an annual demand of 100,000 units. While a weighted average demand could reflect long-term uncertainty, we chose 100,000 units to remain consistent with the financial forecasts driving our revenue projections for Year 1. Using this demand figure, we calculated each supplier's optimal order quantities (EOQ), reorder points (ROP), safety stocks, and total annual inventory costs, including purchase, setup, and holding costs which are summarized in the table below.

To maintain a 98% cycle service level, we applied a Z-score of 2.05 to calculate the safety stock required for each supplier. Due to its short 5-day lead time, M2 requires considerably less safety stock and a much lower reorder point than P3, which operates with a 45-day lead time. These lead time differences directly affect inventory carrying costs—P3 must hold significantly more buffer inventory, resulting in much higher annual holding costs.

M2 Supplier	P3 Supplier
ROP	ROP
2612	19374
EOQ	EOQ
1248	1116
Safety Stock	Safety Stock
689	2067
TAHC (holding cost)	TAHC (holding cost)
\$3,370.96	\$6,739.03
TASC (setup cost)	TASC (setup cost)
\$1,602.34	\$1,433.18
TAPC (purchase cost)	TAPC (purchase cost)
\$1,027,000.00	\$1,027,000.00
TC (total cost)	TC (total cost)
\$1,031,973.30	\$1,035,172.21
M2 is the cheaper option by:	\$3,198.91

Although P3 offers a slightly lower administrative cost per order (reflected in set-up costs), the savings are minimal and do not offset the additional expense of carrying extra inventory. Both suppliers charge the same unit price, so purchase costs remain equal. When combining all relevant inventory costs—purchase, setup, and holding—M2 emerges as the lower-cost option by \$3,199, offering a clear financial advantage compared to P3.

Supplier Recommendation

Beyond the numbers, relationship and strategic considerations further support choosing M2. GPC currently has a strong relationship with M2, having worked with them for years as their largest customer. This familiarity and reliability reduce risk, particularly during the launch of a new system. In contrast, P3 is a large, nationwide manufacturer that recently underwent a private equity acquisition. While they are interested in taking on GPC's circuit board business, the company would likely be a small client in their portfolio, limiting responsiveness and flexibility.

Given M2's cost advantage, shorter lead times, and strong existing relationship, they are the clear choice for the first year of production. Their consistency and lower operational risk will help ensure a smooth and cost-effective launch of the new system.

Quality Management

As GPC transitions from its established commercial lawn care roots into the dynamic consumer gardening market, its philosophy on quality must evolve to meet new demands. The company has historically thrived by delivering high-quality products and strong customer support, earning a trusted reputation in the commercial sector. However, the introduction of the technologically advanced, subscription-based PlantLuv System brings a new level of complexity and customer expectation. In this environment, traditional quality control practices are no longer sufficient; GPC must adopt a modern, integrated, and customer-focused quality management system grounded in Lean, Six Sigma, and other proactive methodologies to ensure success.

GPC's current quality practices are commendable. The use of Statistical Process Control (SPC) charts, Pareto analyses, 100% inspection of finished goods, and supplier ISO 9000 certification all reflect a strong commitment to product integrity. These measures, however, largely represent a reactive approach: finding and fixing defects after they occur. In the consumer market, especially with a product as personalized and tech-dependent as the PlantLuv System, quality must be built into every step of the process. This shift from Quality Control (QC) to Quality Assurance (QA) is critical for maintaining consistency, building trust, and meeting the higher expectations of home gardening consumers.

The Strategic Shift

A strategic improvement GPC should make is implementing Lean principles across the organization. Lean emphasizes the elimination of waste and the optimization of value from the customer's perspective. Tools like Value Stream Mapping can be used to visualize and streamline the entire production process (from soil test creation to spike fulfillment and delivery), highlighting inefficiencies such as excess inventory or redundant steps. Additionally, implementing a 5S system (Sort, Set in Order, Shine, Standardize, Sustain) can enhance organization and minimize variation on the factory floor, which is crucial for maintaining high product quality. Error-proofing methods such as Poka-Yoke should also be introduced to reduce human error and reinforce process reliability. The benefits of Lean extend beyond cost savings; they help create a workplace culture focused on discipline, accountability, and precision.

Simultaneously, the company should embed Six Sigma practices into its operations. Six Sigma's data-driven approach to reducing variability is critical for the PlantLuv System, which relies on sensor accuracy, soil chemistry precision, and consistent user experience. By launching DMAIC (Define, Measure, Analyze, Improve, Control) projects targeting key quality concerns, GPC can address root causes rather than symptoms. Enhancing current SPC usage to include process capability metrics (C_p and C_{pk}) will provide a clearer picture of how well the processes are performing relative to specifications. This level of control will also support scalability as the product line grows, minimizing defects and warranty claims.

Supplier quality management also plays a crucial role in ensuring final product reliability. GPC must move beyond certifications and audits to create more integrated partnerships with suppliers, particularly those providing electronic components. This can include real-time SPC dashboards, supplier performance scorecards, and shared improvement initiatives to align both parties' quality goals. With the circuit board and app components critical to product functionality, these proactive measures will reduce risk and strengthen the supply chain.

Equally important is the application of Quality Function Deployment (QFD), a method for translating customer needs into product features. By using tools like the House of Quality, GPC can align engineering specifications with what customers value - such as ease of use, aesthetic appeal, and real-time feedback. In tandem, Failure Mode and Effects Analysis (FMEA) should be used to anticipate potential failure points

during both design and production, allowing the team to prioritize and address the most critical risks before they impact customers. These techniques promote strategic thinking and ensure product development is aligned with customer expectations from the outset.

Building a Culture of Improvement

Utilizing customer feedback will be essential; the PlantLuv System's app and web platform offer a useful chance to gather data on user experience, system problems, and satisfaction levels in real time. A closed-loop quality improvement procedure ought to include this input. This dedication will be strengthened throughout the company by promoting a culture of quality ownership among staff members, backed by training and the ability to take action on quality-related issues. Quality-focused workers are more likely to maintain high standards on the manufacturing line and offer innovative solutions. GPC has to adopt a proactive, system-wide concept based on Lean and Six Sigma in place of its current reactive, traditional quality strategy. Employee empowerment, supplier collaboration, and ongoing feedback should all be a part of this new strategy. With these enhancements, the business will be in a strong position to provide a dependable, superior PlantLuv System that satisfies customer demands and fosters enduring brand loyalty in the cutthroat home gardening industry.

Operations as a Value-Added Function

In the strategic architecture of GPC, the operations function extends beyond its traditional role as a cost center or back-office support mechanism. As the company prepares to navigate the pivotal launch of the PlantLuv System, its operations division emerges as its most vital and potent strategic weapon. This powerful engine will translate an innovative vision into tangible, defensible customer value. While marketing crafts the brand promise and finance allocates the necessary resources, it is the operations function that must execute the corporate strategy with precision, agility, and foresight. It is within the complex domain of operations (encompassing process design, supply chain management, quality assurance, and project execution) that GPC's competitive advantage will be forged, sustained, and defended. The firm's ability to deliver on the unique value proposition of the PlantLuv System is wholly dependent on the strategic management of its operational

capabilities, which collectively drive market-defining innovation, guarantee unwavering quality and brand trust, and architect a competitive and profitable cost structure.

Driving Innovation and Differentiation

Operations serve as the primary driver of innovation and strategic differentiation. The PlantLuv System is a concept built on technological advancement and deep personalization, and it is the operations function that gives this concept its physical form, logistical feasibility, and functional excellence. The system's core differentiator? Its ability to provide tailored plant spikes based on an individual customer's soil analysis: fundamentally an operational feat. This model represents a seismic shift from the company's historical paradigm of mass-producing standardized commercial equipment. It requires the design and implementation of a sophisticated, data-driven process that can seamlessly convert a unique data point (a soil test result) into a specific, packaged, and fulfilled order. This is not mass production; it is mass customization, a capability that demands an agile and intelligent operational backend capable of managing immense complexity. By leveraging flexible manufacturing systems, efficient batch scheduling, and customizable labeling, the operations team creates a formidable competitive advantage that is exceptionally difficult and costly for rivals to replicate.

This innovative capability is directly shaped by strategic operational decisions, most notably the choice of production technology. The consideration of a worker-paced versus a machine-paced assembly system extends far beyond a simple cost-benefit analysis; it is a decision that fundamentally defines the product's quality, consistency, and scalability, thereby shaping its brand identity. Opting for the more automated, machine-paced process, though requiring a greater initial investment of \$750,000, directly supports a value proposition centered on a high-tech, reliable, and precisely manufactured "smart system." The resulting lower variable cost of \$1.12 per unit, compared to \$4.08 for the manual line, is a significant financial benefit, but the strategic value lies in the consistency and precision that automation provides. This operational choice is, in effect, a marketing decision, reinforcing the brand's promise of scientific accuracy and technological sophistication. Furthermore, this operational prowess is the launchpad for all future innovation. Buck Bridges' long-term vision of incorporating internet-connected sensors and fully automated gardening systems is entirely

contingent on the firm's operational and engineering capabilities. The ability to design manufacturability, develop robust prototypes, and scale production for these future technologies resides within the operations function. In this sense, operational excellence today is a direct and necessary investment in the market leadership of tomorrow, positioning operations as a forward-looking, strategic growth driver.

Guardian of Quality and Brand Trust

Concurrent with driving innovation, the operations function stands as the vigilant guardian of quality and reliability, the twin pillars upon which brand reputation and customer trust are built. In the consumer market, particularly for a premium, subscription-based product, the stakes for quality are exponentially higher than in the commercial sector. A product failure in the B2B world might be rectified by a service call from a trusted sales representative. In the B2C world, a single failure—such as a bad batch of spikes that damages a customer's prized plants—can trigger a cascade of negative online reviews, social media backlash, and irreparable harm to the brand. Operations is the primary defense against this existential risk. By embedding Lean and Six Sigma methodologies throughout the value chain, operations ensure that quality is not a mere checkpoint but a foundational principle. The use of tools such as 5S, Value Stream Mapping, and DMAIC allows the company to maintain the highest product standards while scaling production.

The effectiveness of the PlantLuv System hinges on the unerring accuracy of its diagnostic components and the precise chemical composition of its nutrient spikes. It is the operations function, through the disciplined application of Statistical Process Control (SPC), a comprehensive quality assurance framework, and stringent supplier management, that guarantees this essential level of performance. The strategic selection and management of suppliers is a clear example of this operational mandate. The choice between Mona Manufacturing (M2) and Pine Peak Products (P3) is not merely tactical. While P3's lower ordering cost of \$16 is attractive, M2's drastically shorter lead time of 5 days (versus 45 days for P3) provides a profound strategic advantage. This operational agility allows for greater responsiveness to demand fluctuations, requires significantly less capital to be tied up in safety stock to maintain the target 98% service level, and reduces the risk of stockouts that would be fatal for a subscription service built on the promise of timely delivery. This reliability, engineered by operations, is a core and tangible part of the customer's perceived value.

Furthermore, the aggressive 29-week launch timeline is an operational challenge that, if met, confers a significant strategic advantage. The ability to use project management techniques to execute this plan successfully allows GPC to capture first-mover advantage, build market momentum, and establish the brand before competitors can fully react.

Cost Structure

Ultimately, operations is the architect of the company's cost structure, which determines the product's market pricing, consumer accessibility, and eventual profitability, all the while providing innovative products with unwavering quality. Every dollar saved through operational efficiency is a strategic resource that may be used for marketing initiatives, research & development, or providing better value to the customer. The decision to adopt the machine-paced production system, for example, creates a variable cost advantage of \$2.96 per unit. Over the initial forecast of 100,000 units, this single operational decision yields a \$296,000 improvement in contribution margin in the first year alone, fundamentally altering the project's financial viability and its Net Present Value. In its most basic form, this is value generation. Managing the intricacies of a new distribution model is another way in which financial management is demonstrated. Operations must acquire new skills in last-mile logistics, inventory management across channels, and e-commerce fulfillment in order to transition to an omnichannel strategy. Operations makes sure that this intricate fulfillment network is both effective and economical by strategically examining third-party logistics (3PL) partnerships, optimizing warehouse locations, and putting modern inventory systems into place. This directly protects the subscription model's profitability.

Cross-Functional Synergy

In addition to these fundamental foundations, the operations team is essential to fostering cross-functional cooperation and organizational alignment. Operations must serve as the strategic nerve center of the business; it cannot operate in a vacuum. For GPC, this entails setting up reliable, instantaneous departmental feedback channels. For example, to guarantee product availability and avoid stockouts throughout a campaign, marketing's promotional calendar needs to be closely informed by operational capacity data and manufacturing schedules. On the other hand, in order to initiate root-cause analysis and process enhancements, customer

service input concerning shipping problems or component usability needs to be directed right away to the operations and engineering teams. The corporation is transformed from a collection of separate departments into a single, responsive, and learning organization through this systemic integration, where operational data informs marketing and consumer data inspires operational improvement.

At GPC, the operations function is the glue that keeps the whole strategic endeavor together, not just a part of the corporate strategy. It is the cohesive, dynamic force that transforms creative ideas into superior products, oversees the intricate international supply chains that reliably deliver them, and carefully regulates the cost structures that guarantee their sustained profitability. The PlantLuv System's ultimate success will be determined by how well it is implemented, not how cleverly it was conceived. GPC's most powerful and essential strategic tool for securing the future of home gardening is execution, which is the exclusive area of operations.

Financial Evaluation and Risk Analysis

Summary of Key Assumptions

Our baseline analysis incorporates several foundational assumptions; each grounded in practical business logic and aligned with capital budgeting principles. For revenue projections, we used a \$35/unit net wholesale price rather than the \$75 retail price. This reflects the actual cash inflow to GPC after accounting for channel costs and retailer markups. Using the net price enables us to pair revenue with more accurate cost of goods sold estimates, yielding a cleaner measure of contribution margin. We excluded prior R&D and packaging expenditures from our model as they represent sunk costs that are irrelevant to future cash flows. We assumed Machine E will be purchased in Year 4, coinciding with the first year that projected unit volume exceeds the 125,000-unit capacity of Machines A–D. Given its 5-year depreciable life and immediate deployment, depreciation begins in Year 4 and continues through Year 8. To ensure uninterrupted capacity in Years 9 and 10, we assume the machine is repurchased in Year 9, initiating a new depreciation cycle. Any residual book value at the end of Year 10, though unrecovered, could be reallocated to future projects, supporting the strategic reuse of long-term assets. We elected to include a \$1,000 annual remediation cost in Years 1–9 to conservatively account for compliance obligations and long-term environmental risks. While the case presents multiple options—including ignoring the cost entirely or applying a \$300,000 outflow every three years—we opted for a smoothed annual charge to avoid distorting individual year cash flows. We excluded Year 10 not because we assume no risk, but because under the every-three-year structure, no remediation expense would fall in that final year of the project cycle. We excluded the \$90,000 annual interest expense from our cash flow projections because financing decisions should be evaluated separately from investing decisions, including it in operating cash flows would misstate the project’s financial performance. We accounted for erosion risk from cannibalization of an existing product line, but the new project’s net cash flows are substantially higher, easily offsetting the impact. Over the 10-year period, the average net cash flow from erosion effects is approximately \$4.87 million positive, highlighting the extent to which the new product compensates for lost sales. Lastly, we treated the initial capital investment as an opportunity cost by benchmarking it against returns from

comparably risky investments, such as a mutual fund. This ensures that only projects generating returns more than the firm's required rate of return are considered viable, in line with shareholder value maximization.

Recommendation

We recommend that GPC move forward with the new product launch. Using a project-specific cost of capital of 8.21%, the project generates an NPV of \$10.32 million and an IRR of 180%, indicating exceptional financial performance and value creation. Even when factoring in the initial investment of \$673,424, the purchase and repurchase of machinery over the project's life, and the erosion of existing product sales, the projected cash flows remain strong. The project's profitability holds under several conservative assumptions, including the addition of a 1.2% liquidity risk premium and the inclusion of ongoing remediation costs to account for regulatory risk.

Cost of Capital Rationale

To estimate the project's cost of capital, we began by calculating GPC's levered beta through a regression of firm returns against the S&P 500. We found a levered beta of 1.040, which we then unlevered using the firm's average debt-to-equity ratio and tax rate from the past five years from Exhibit F2, resulting in an unlevered beta of 0.896. To tailor the beta to this specific project, we developed a risk scaler methodology. We chose to use gross margin (GM) data rather than revenue because GM captures the profitability after direct costs, making it a better reflection of a project's true financial sensitivity. However, since market data is expressed in percentages rather than dollar values, we first needed to convert GM into a return-based format using the equation: $GM\ Return_t = (GM_t - GM_{t-1}) / GM_{t-1}$. This equation reflects year-over-year gross margin growth, giving us a format compatible for direct comparison with market returns in regression analysis. We regressed the GM returns of each individual project against historical market returns to compute the project-level GM-to-market return slopes. Then, we calculated the risk scaler by dividing each individual project's slope by the slope of the aggregate GM return (i.e., the sum of all five projects' gross margins) regressed against those same market returns. This ratio reflects how much more or less sensitive each project is compared to the firm overall. While most projects yielded risk scalars within a reasonable range, Project Q's scalars were negative,

likely due to low correlation or an inverse relationship with market trends. Since our proposed project is internally viewed as most similar to Project R, we used Project R's risk scaler of 1.024. Applying this to the firm's unlevered beta gave us a project beta of 0.917.

To complete the CAPM calculation, we derived a current risk-free rate of 3.64% using the 10-year Treasury price and coupon rate from Exhibit F4 in Excel's =RATE function, adjusting for semi-annual coupon payments. This matches the 10-year duration of our current project. For the market risk premium, we used the difference between historical market returns and 1-year T-bill returns, providing a long-run average premium of 3.67%. Following the CAPM formula, we found the project cost of capital = $3.64\% + (0.917 \times 3.67\%) = 7.01\%$. Finally, we added a liquidity risk premium of 1.2%, bringing our final cost of capital to 8.21%, with further justification for this premium discussed in the next section.

Treatment of Risk Premiums

In addition to the baseline CAPM estimate, we considered two risk premiums: a liquidity risk premium and a default risk premium. After evaluating the nature of the investment, we decided to include a 1.2% liquidity risk premium. This adjustment reflects the fact that the capital tied up in this internal project is not easily liquidated, especially over the 10-year horizon. Unlike publicly traded securities, there is no ready secondary market for recovering the invested capital, which justifies a higher required return to compensate for the opportunity cost of limited liquidity. However, we chose to exclude the 2.4% default risk premium. While there is some concern among team members about the potential for missed dividend payments during the early phases of the project, our analysis shows that the project generates positive cash flows every year following the initial investment, reinforcing its ability to meet obligations. Including this premium would likely overstate the project's risk and unfairly inflate the discount rate, potentially distorting the investment's valuation.

Sensitivity Insights

To evaluate the impact of demand uncertainty, we conducted a sensitivity analysis based on the probabilistic scenarios outlined by the operations team. The baseline scenario that we have been evaluating, occurring with

a 55% probability, assumes 100,000 units sold in Year 1. As previously established, this leads to the purchase of Machine E in Year 4 and its repurchase in Year 9 to maintain production, yielding an NPV of \$10.32 million and an IRR of 180%. The pessimistic scenario assumes Year 1 demand is only 50,000 units, with a 10% probability. Demand never exceeds the 125,000-unit capacity of Machines A–D, so Machine E is never purchased. This results in significantly lower capital expenditures. Nonetheless, the project still generates a strong NPV of \$4.49 million and an IRR of 107%, illustrating profitability even under unfavorable market conditions. The optimistic scenario, which carries a 35% probability, assumes demand begins at 175,000 units in Year 1. To meet this volume, Machine E must be added immediately—either purchased in Year 0 if anticipated, or in Year 1 and placed into service right away. This upfront investment drives an NPV of \$21.47 million and an IRR of 202%, highlighting the project’s strong upside potential. Considering the weighted probabilities of each case, the expected NPV across all three demand scenarios is approximately \$13.64 million. These results affirm the project’s resilience and attractiveness across a range of plausible market outcomes.

Strategic Implications

This project presents strong strategic benefits for GPC, particularly through its investment in automation. The addition of Machine E expands production capacity beyond the limitations of the manual system and reduces variable costs by limiting reliance on labor. This shift enables more scalable and efficient operations as demand grows. The model’s performance across multiple demand scenarios demonstrates the project’s ability to adapt to varying market conditions, supporting the company’s long-term goal of sustainable, cost-effective growth.

Alternative Scenario Analysis & Implications

In business, scenarios are constantly evolving proving no financial decision to be a static relationship. This is no different for GPC, and to provide a holistic and stable recommendation, an alternate scenario was considered to ensure our recommendation can withstand evolving situations.

More specifically, the alternative scenario adjusted details in capital investments. This impacted equipment E that, if purchased, equipment E would qualify for immediate bonus depreciation. Year 4 and 9 both receive these outflows of \$750 thousand, instead of \$150 thousand, as this is when we initially purchase equipment E in year 4. This is due to the quantity exceeding 125,000 units, and then again in year 9 as the 5-year service life of the equipment will be expired and need replaced to maintain their level of production.

Equipment D also faces some alterations as in the alternate scenario it is sold for \$200 thousand, representing a gain on its original estimated salvage value of \$30 thousand. Taking in the effects of tax on the gain earned from the difference between the market value of \$200 thousand and the book value of \$30 thousand, the alternate net salvage value on equipment D is \$161.138 thousand. This will be a cash flow in both years 5 and 10 due to the 5-year service life and the initial purchase of equipment D in year 0 (with consumption beginning year 1).

A downside scenario was also considered in this alternative situation. Competitive forces caused retail prices to shift from \$75 a unit to \$55 a unit. Channel and promotional costs fell to \$31 from \$40, but the new wholesale price fell to \$24 a unit. Following the assumptions set for the initial scenario to ensure consistency, the revenue was calculated with the wholesale price, not retail revenue, to reflect the curate inflow to GPC without the retailer markups to provide an accurate, non-inflated, cash flow. A final change occurred when a \$1 per unit improvement in material costs in comparison to the baseline, decreasing the per unit expense.

Considering these four adaptations, year 0 cash flows remain consistent, but beginning year 1 cash flows decrease greatly. The table below shows these drastic decreases in annual cash flows from the initial scenario to the alternate. These four changes may seem minor, but their

Year	Baseline Cash Flow	Alternative Cash Flow	Percent Change
0	\$ (673,424)	\$ (673,424)	0%
1	\$ 1,301,958	\$ 540,415	-58.49%
2	\$ 1,018,831	\$ 303,611	-70.20%
3	\$ 1,139,677	\$ 354,218	-68.92%
4	\$ 821,215	\$ 95,820	-88.33%
5	\$ 1,368,699	\$ 525,229	-61.63%
6	\$ 1,957,798	\$ 876,622	-55.22%
7	\$ 2,168,095	\$ 984,302	-54.60%
8	\$ 2,397,829	\$ 1,101,414	-54.07%
9	\$ 3,398,759	\$ 650,197	-80.87%
10	\$ 2,390,976	\$ 1,723,324	-27.92%

implications on annual cash streams proved to be drastic. Overall, this alternative scenario caused a drastic **loss** in total annual cash flows of \$10,808,685 or 62.51% of the original cash. Furthermore, in the traditional scenario the Net Present Value was \$10,323,311.06. However, in the alternative scenario the Net Present

Value decreased to \$3,590,804. Additionally, the initial internal rate of return resulted at 180%, and the alternative scenario resulted with a 66.09% internal rate of return.

While the alternative scenario was able to remain positive, the extent of its earnings is not consistent to the original scenario after their introduction. This raises some alarms as to the stability of this investment.

Furthermore, this heightens concern as all these changes that lead to the loss are unpredictable and uncontrollable by GPC. GPC cannot directly control the laws surrounding the depreciation of equipment E, they cannot control the market values and predict for gains such as the one on equipment D, they cannot know the true market trajectory to predict the price drops down to \$24 a unit, and they cannot rely on a future dip in variable costs. The result of all these unpredictable changes that can accelerate a decline in profit makes the project riskier and more susceptible to influence. This is something we urge management to consider the effects on profitability to determine whether these unknowns are worthy of challenging.

Ultimately, this project in both scenarios still proved high levels of profitability. While unpredictability caused the steep decline in return between the two scenarios, we still recommend continuing with the introduction of the PlantLuv project. However, it is recommended that management pays close attention to any shifts in the market, political climate, and any related expense to consistently monitor the project's returns. With careful monitoring, GPC can prohibit any extreme declines into negative net present values or internal rate of returns to stay ahead of any major financial disasters. Contingent on this, we urge the project to begin.